

ARPITA PATNAIK

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Professional Summary

Results-driven Software Engineer with 5 years of experience specializing in backend development, database optimization, and full-stack web application development. I am experienced in writing production-level code using Python and Django, and have extensive experience working with databases such as PostgreSQL and Redis to deliver robust and scalable solutions.

Education

University of Florida, Gainesville

Master of Science in Computer Science

Aug 2023 – May 2025

GPA: 3.63/4

National Institute of Science and Technology, India

Bachelor of Technology in Information Technology

Aug 2012 – 2016

CGPA: 8.19/10

Experience

Mindtree Ltd. (Bengaluru, India) | **Senior Software Engineer**

Jul 2016 – 2021

- Reduced P90 latency of our critical user service API by 75% (from 20 to 5 seconds) using a read-through cache (Redis) and a smart retry mechanism, improving system performance and stability.
- Built a feature fencing system with Admin dashboard in Django/Redis that enabled real-time feature control and group-based targeting, reducing downtime risks and enabling faster product iterations.
- Engineered auto-update component for desktop applications using C Sharp and automated deployment processes with CI/CD pipeline implementation and API integration.
- Took ownership of adding unit testing strategies using Xunit and TypeMock, achieving 75% coverage in one quarter following software development lifecycle standards.
- Revamped user details section to aggregate data from multiple reports, improving system architecture, database management, and user experience.
- Designed and implemented theme/skin using DevExpress, WinForms, and WPF for five Windows applications with Dark Mode UI, enhancing full stack development capabilities.

Projects

M3: Mood Movie Matcher | *Python, PyTorch, Machine Learning, NLP, scikit-learn, Transformers* | Github Repo

- Developed ML-powered movie recommendation system using RoBERTa sentiment analysis and Autoencoders on MovieLens dataset, enabling discovery of personalized best matches and hidden gem films through interactive command-line interface.

Safe Data Cloak and Lift The Veil | *Python, SpaCy, scikit-learn, sentence transformers* | Github Repo

- Designed scalable data privacy system utilizing NLP transformers for precise redaction of sensitive data including PII, dates, phone numbers, and domain-specific concepts.
- Built unredaction functionality using NLTK and Random Forest algorithms with enterprise API integration capabilities.

Deal Hunter Ai - Multi-Agent Deal Intelligence System | *Python, Llama 3.1, OpenAI, RAG, Modal.com, Gradio* | Github Repo

- Architected production LLM-based AI system with 9 specialized agents orchestrating fine-tuned Llama 3.1-8B (QLoRA), RAG pipeline (400k embeddings), and ensemble ML for autonomous deal discovery and accurate price prediction.
- Deployed scalable GPU inference on Modal.com achieving sub-2s latency with real-time notifications, demonstrating end-to-end MLOps from model fine-tuning to production deployment and distributed systems design.

LeoZen Bot | *Python, Discord API, Web Scraping, zenquotes.io* | Github Repo

- Developed automated Discord bot with API integration for user engagement and motivational quote delivery, extracting quotes from zenquotes.io using web scraping techniques.

Technical Skills

Languages: Python, C#, JavaScript, TypeScript, HTML/CSS, SQL

Frameworks: Django, Flask, RESTful APIs, ReactJS

AI and ML: PyTorch, NumPy, Pandas, TensorFlow, Llama, OpenAI API, LLM, Agentic AI

Databases: PostgreSQL, Redis

Cloud and DevOps: AWS, Git, Jenkins, Docker, CI/CD Pipeline, Linux

Testing and Quality: Unit Testing, XUnit, TypeMock, pytest, TDD, Postman, Insomnia

Certifications and Coursework

Certifications: AWS Certified Cloud Practitioner (April 2025), Meta Back-End Developer

Coursework: Machine Learning, Analysis of Algorithms, Distributed Operating Systems, Computer Networks, Data Engineering, Computer Network Security

Leadership and Extracurricular

- Received Achievement Award Scholarship by University of Florida for academic excellence and software engineering achievements.
- Teaching Assistant for Analog Electronic Circuits, guiding 50+ undergraduates in circuit design, analysis, and technical problem-solving.
- Directed coordination and logistics for 50+ large-scale events at the O'Connell Center, some hosting 10,000+ attendees, ensuring flawless execution, operational efficiency, and exceptional participant experiences through strategic planning and team leadership.